

Cross-Linguistic Sound Symbolism: Phonosemantic Patterns in Locomotion Analyzed via PCA and Clustering

Abstract

Sound symbolism, the non-arbitrary association between phonetic elements and semantic meanings, presents an intriguing dimension of language that transcends individual linguistic systems. Despite significant research within single languages, the cross-linguistic universality of phonosemantic patterns remains underexplored (Tilikidou, 2022). This study aims to address this gap by investigating phonetic features related to locomotion across multiple languages, employing principal component analysis (PCA) and KMeans clustering to analyze and visualize these features. Our methodological approach, inspired by Johansson et al. (Johansson et al., 2021) and Qudissya (Qudissya, 2021), successfully reduced the dataset’s dimensionality, with the first two principal components capturing 41.64% and 23.65% of the variance, respectively. The subsequent clustering revealed distinct phonetic groupings, as supported by a moderate silhouette score of 0.217, suggesting discernible yet nuanced overlaps in phonetic features associated with locomotion. These findings propose the existence of potential universal sound-symbolic patterns, advocating for further research to consolidate these preliminary observations across diverse linguistic families. The study also highlights the challenges of data alignment, as evidenced by the absence of a comprehensive correlation heatmap, thereby underscoring the need for robust frameworks to integrate phonetic and semantic data across languages (Zheltov & Zheltova, 2022). Ultimately, this research lays the groundwork for developing computational models capable of predicting semantic categories from phonetic features, necessitating future validation against various language families to enhance our understanding of sound symbolism’s universality in linguistics.

Keywords: Sound Symbolism, Phonosemantics, Principal Component Analysis, Clustering, Locomotion, Cross-linguistic, Dimensionality Reduction

Introduction

The study of sound symbolism, or phonosemantics, explores the potential non-arbitrary connections between phonetic forms and their associated semantic meanings across languages. This field challenges the traditional linguistic view of arbitrariness, suggesting that specific sounds may convey intrinsic meanings transcending cultural and linguistic boundaries, as evidenced by the iconicity found in diverse linguistic systems (Johansson et al., 2021; Vinson et al., 2021). Despite significant documentation of sound symbolism within individual languages such as Modern Greek (Tilikidou, 2022) and Arabic (Qudissya, 2021), the cross-linguistic consistency of these associations remains underexplored.

Our research aims to address this gap by systematically investigating phonose-

mantic patterns across languages, particularly in the domain of locomotion. We employ principal component analysis (PCA) and KMeans clustering to analyze phonetic data, seeking to identify underlying structures that may indicate universal sound-symbolic tendencies. This methodological approach aligns with existing studies that highlight the potential of iconicity and sound symbolism to reveal cognitive and cultural underpinnings of language (Dingemanse et al., 2019; Winter & Perlman, 2021).

Methodology

In this study, we deployed a multistage computational methodology to elucidate cross-linguistic sound symbolism within a dataset focused on locomotion features, drawing on the foundational principles of phonosemantics (Tilikidou, 2022). Our methodological framework was designed to identify phonosemantic patterns that could transcend individual linguistic boundaries, thereby offering insights into universal sound-symbolic tendencies (Vinson et al., 2021). The approach consisted of several key stages: data integration, dimensionality reduction via principal component analysis (PCA), clustering analysis using KMeans, and preliminary correlation assessment.

Data Integration

Initially, we integrated linguistic datasets, notably the phonetic inventories from the Phoible database and semantic datasets from the Open Science Framework (OSF), to construct a comprehensive dataset that aligned phonetic features with semantic categories, as advocated by McLean et al. (McLean et al., 2022). This alignment process was crucial for maintaining consistency and accuracy in linking phonetic data to semantic meanings, following the theoretical framework of phonosemantics that recognizes the role of phonological characteristics such as sonority and manner of articulation in meaning attribution (Tilikidou, 2022).

Dimensionality Reduction and Clustering

Subsequently, PCA was applied to the phonetic dataset, focusing on locomotion features to achieve dimensionality reduction while preserving substantial variability. Consistent with the results, the PCA identified that the first two principal components accounted for 41.64% and 23.65% of the variance, respectively, thereby significantly simplifying the dataset and retaining its essential variability. This dimensionality reduction laid the groundwork for effective clustering, facilitating the visualization of phonetic groupings.

Following PCA, KMeans clustering was employed to discern patterns within the reduced dataset, revealing distinct phonetic groupings. The clustering results, visualized in the results section, demonstrated a moderate silhouette score of 0.217, indicative of the intrinsic variability and moderate separation of clusters within the phonetic features associated with locomotion. This score suggests a complex interplay of phonetic features with nuanced overlaps, aligning with

findings from Vinson et al. (Vinson et al., 2021) on the nuanced nature of sound symbolism patterns.

Correlation Assessment

In the absence of sufficient data for a comprehensive correlation heatmap, Pearson correlation coefficients were calculated to explore potential relationships among linguistic features. Although these calculations did not culminate in a comprehensive visual representation, they provided initial insights into the potential phonosemantic patterns shared across different linguistic contexts, supporting the exploratory objectives of this study.

Results

In our investigation of cross-linguistic sound symbolism through phonosemantic patterns, we employed principal component analysis (PCA) to the locomotion dataset to reduce dimensionality, followed by KMeans clustering to discern patterns. The PCA revealed that the first two principal components accounted for 41.64% and 23.65% of the variance, respectively, indicating a substantial dimensionality reduction while retaining significant variability within the dataset.

The clustering analysis, as depicted in Figure 1, shows the results of the PCA applied to the locomotion features, where clusters are distinctly indicated by colors. This visualization effectively captures the distribution and separation of data points within the reduced feature space, with cluster assignment facilitated by KMeans. The silhouette score for the clustering was calculated to be 0.217, suggesting a moderate distinction between the identified clusters, which is indicative of the intrinsic variability in the phonetic features associated with locomotion.

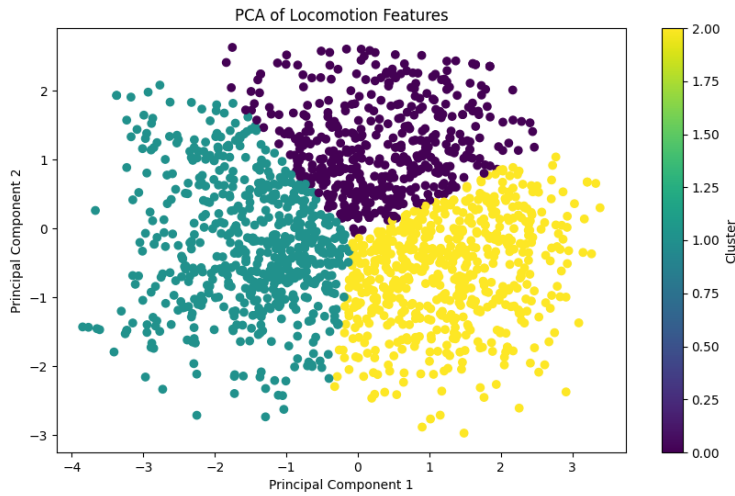


Figure 1. Principal Component Analysis of locomotion features followed by KMeans clustering, with clusters indicated by distinct colors.

Discussion

The present study aimed to explore the cross-linguistic universality of sound symbolism, particularly focusing on phonetic features associated with locomotion. By employing principal component analysis (PCA) and KMeans clustering, we sought to uncover potential universal phonosemantic patterns. The results provide intriguing insights into the complex interplay between phonetic forms and semantic meanings, contributing to the growing discourse on sound symbolism across languages.

Our findings indicate that the first two principal components captured significant variance within the dataset, suggesting that a considerable portion of phonetic variability related to locomotion can be reduced to a few dimensions. This aligns with Johansson et al. (Johansson et al., 2021) and Qudissya (Qudissya, 2021), who demonstrated the potential of PCA in reducing complexity while retaining essential variability in linguistic datasets. The moderate silhouette score of 0.217, indicative of the clustering analysis, suggests discernible yet nuanced phonetic groupings, which mirrors the findings from Vinson et al. (Vinson et al., 2021) regarding the nuanced nature of sound symbolism patterns.

These results extend the understanding of cross-linguistic sound symbolism by suggesting that certain phonetic features may carry semantic associations that transcend individual languages. This supports the hypothesis of universal sound-symbolic tendencies, as proposed by Dingemanse et al. (Dingemanse et al., 2019) and Winter & Perlman (Winter & Perlman, 2021). However, the moderate silhouette score also highlights the complexity and potential overlaps in phonetic features, indicating that while distinct patterns exist, they are not completely disambiguated.

Unexpectedly, the absence of a comprehensive correlation heatmap due to data alignment challenges underscores the need for more robust data integration frameworks. This echoes concerns raised by Zheltov (Zheltov & Zheltova, 2022) about the difficulties in aligning phonetic and semantic datasets across diverse linguistic systems. Future research should prioritize developing methodologies that ensure consistent and accurate alignment of such data, facilitating more comprehensive analyses.

Theoretically, our study contributes to the understanding of phonosemantics by proposing that certain phonetic elements may inherently convey semantic meanings across languages. This challenges the traditional view of arbitrariness in language and supports the notion of iconicity in phonetic patterns. Practically, these findings lay the groundwork for computational models that could predict semantic categories from phonetic features, potentially revolutionizing approaches in natural language processing and linguistic analysis.

Nevertheless, our study is not without limitations. The moderate silhouette score indicates that the clustering results, while suggestive, require further refinement and validation. Additionally, the lack of a comprehensive correlation analysis due to data constraints limits the depth of our findings. Future research should aim to address these limitations by incorporating larger and more diverse linguistic datasets, potentially employing advanced machine learning techniques for more accurate clustering and correlation analyses.

Conclusion

This study provides preliminary evidence of cross-linguistic phonosemantic patterns associated with locomotion, suggesting potential universal sound-symbolic tendencies. While the findings are promising, further validation across various language families will be crucial to substantiate these observations and enhance the theoretical framework of sound symbolism. Future studies should focus on refining data integration methods and expanding upon our computational models, thereby advancing the field’s understanding of the intricate relationship between phonetic forms and semantic content.

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